

Health and Retirement Study 2006



Physical Measures & Biomarkers Booklet

IWER: Text in ***bold and italics*** is to be read to the respondent. Normal text is interviewer instructions or questions that are to be answered by you.

IWER: Mark an 'X' by each measurement for which R is eligible.

- _____ Blood Pressure
- _____ Breathing
- _____ Hand Strength
- _____ Balance Tests (with 30 second full-tandem) **Record this on the top of pages 11 & 12**
- _____ Balance Tests (with 60 second full-tandem) **Record this on the top of pages 11 & 12**
- _____ Walking Test **Record eligibility at the top of pages 13 & 14**
- _____ Height
- _____ Weight **Record eligibility at the top of pages 17 & 18**
- _____ Waist
- _____ Saliva
- _____ Blood Spot Assays

IWER: Before beginning the physical measurements, you must have a signed consent form from the Respondent.

HEALTH AND RETIREMENT STUDY 2006

PHYSICAL MEASURES CONSENT FORM

I have already agreed to participate in the Health and Retirement Study funded by the National Institute on Aging and conducted by the University of Michigan. The study will help provide a very complete picture of how people 50 years of age and older living in the United States are faring as they head into retirement and enjoy their retirement years.

In addition to completing the questionnaire, I am now being asked to complete some physical measurements. The physical measures conducted will allow researchers to better understand the connections between health status and other indicators of interest such as economic and employment status.

I understand that if I agree to participate it means the following:

- a) I understand that I may be asked to complete up to 8 different physical measurements which involve standing, walking, exhaling, gripping an object with my hands and having my blood pressure, height, weight, and waist measurements taken.
- b) I understand that I will complete these measurements only if I agree to do so. My participation in this research is completely voluntary. I am NOT required to do this, and I can stop my participation at any time.
- c) There is no additional incentive for completing these measures, nor is there a penalty if I choose not to complete this component.
- d) The results of the physical measurements will be kept strictly confidential in the same way as the rest of the interview data. I will not be identified in any reports on this study. Records will be kept confidential to the extent provided by federal, state, and local law. However, the Institutional Review Board, the sponsor of the study (i.e. NIH, FDA, etc.), or university and government officials responsible for monitoring this study may inspect these records.
- e) A Certificate of Confidentiality has been obtained from the Department of Health and Human Services for this study. With this Certificate, the investigators cannot be forced (for example by court subpoena) to disclose information that may identify me in any federal, state, or local civil, criminal, administrative, legislative, or other proceedings. Disclosure will be necessary, however, upon request of DHHS for audit or program evaluation purposes.

TURN OVER

- f) I will be sent a copy of my blood pressure results as well as an indication if it is above the normal range. I will be instructed to share this information with my doctor.
- g) It will take approximately thirty (30) minutes to complete these physical measures.
- h) There are no known risks associated with doing this interview or with participation in this study. In the unlikely event of any injury resulting from the research, no reimbursement, compensation, or free medical treatment is offered by any of the co-sponsors of this research project.
- i) I understand that although I may not receive direct benefit from my participation, others may ultimately benefit from the knowledge obtained in this study.
- j) I may ask questions about this procedure at any time and can expect truthful answers. I can ask the interviewer or contact the Survey Director, Heidi Guyer, at the University of Michigan: (800) 759-7947.
- k) I will receive a copy of this form.

Please print name: _____

Please sign here and date: _____ **Date:** _____

Interviewer's signature: _____ **Date:** _____

Should you have questions regarding your rights as a research participant, please contact the Health Sciences Institutional Review Board, Kate Keever, 540 East Liberty Street, Suite 202, Ann Arbor, MI, 48104-2210, 734-936-0933, email: irbhsbs@umich.edu.

IWER: Did the R sign the Physical Measures consent form? (Circle one)

1 YES - Consent signed → continue to Blood Pressure Measurement (page 1)

5 NO - Consent not signed → Do not complete these measures. Skip to saliva consent instructions on page 21.

BLOOD PRESSURE

Equipment needed: Omron HEM-780N Monitor, Batteries, Stop Watch

“Now let’s talk about the first activity. I’d like to measure your blood pressure using this monitor and cuff which I will secure around your left arm. I would like to take three blood pressure measures. I will ask you to relax and remain seated and quiet during the measurements. First, I will place the cuff on your left arm. Once the cuff is placed appropriately on your arm and we are ready to begin, I’ll ask you to lay your arm on a flat surface palm facing up so that the center of your upper arm is at the same height as your heart. I will then press the Start button. The cuff will inflate and deflate automatically. After we have completed all three measures, I will give you your results.”

IWER: Demonstrate the test.

“Do you have a rash, a cast, edema (swelling) in the arm, open sores or wounds, or a significant bruise where the blood pressure cuff will be in contact?” (IWER: Circle one)

1 YES → do not complete this measure and answer question in box below

5 NO → continue

“Do you understand the directions for this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

IWER: Why didn’t R complete the blood pressure measurement? (Circle all that apply)

1.....R felt it would not be safe

2.....IWER felt it would not be safe

3.....R refused or was not willing to complete the test

4.....R tried but was unable to complete test

5.....R did not understand the instructions

6.....R had a rash, a cast, edema, open sores or wounds, or significant bruise where the blood pressure cuff would contact R’s arm

97.....Other (Specify): _____

→ Go To Next Measurement

INSTRUCTIONS FOR ADMINISTERING BLOOD PRESSURE TEST:

- 1) Insert arm cuff plug into jack on the side of the monitor.
- 2) Instruct R to remove bulky clothing from upper left arm.
- 3) Instruct R to sit in a chair with his/her feet flat on the floor and place his/her upper arm on a table.
- 4) Place the cuff on the R's left arm approximately ½" above the elbow. Position the blue marker over the brachial artery on the inside of the arm.
- 5) Press the sewn hook material firmly against the side of the cuff. [The cuff should make direct contact with the R's skin]. The R should easily be able to fit his/her index finger between the cuff and the arm.
- 6) Instruct the R to rest his/her arm comfortably on a support (like a table) with palm facing upward so the cuff is at the same level as the heart.
- 7) Press the START/STOP button and instruct the R to remain still. The cuff will begin to inflate automatically. Numbers will appear on the display.
- 8) The cuff will then automatically deflate. The monitor will first display the SYSTOLIC and DIASTOLIC readings. RECORD time of reading and both SYSTOLIC and DIASTOLIC readings in chart below. Record the pulse identified by the letter P.
- 9) Press the START/STOP button to repeat the measure. The cuff does not need to be removed or loosened in between readings. If you receive an error message, remove cuff, reposition and try again.
- 10) Allow approximately 45 seconds to 1 minute between readings and repeat steps 4-9 two more times.

IWER: Record measurements in chart: (Enter 993 in first systolic reading if R tried but was unable to do it. Enter 999 if R chose not to do it.) If the lowest reading obtained is greater than 160 systolic or greater than 110 diastolic, fill out the High-Blood Pressure Card and leave it with the respondent.

Measurement #	Time of Reading	Systolic Reading	Diastolic Reading	Pulse
1	__ __: __ __ am/pm	mmHg	mmHg	P
2	__ __: __ __ am/pm	mmHg	mmHg	P
3	__ __: __ __ am/pm	mmHg	mmHg	P

IWER: Which arm was used to conduct the measurements? (Circle one)

- 1..... Left arm
- 2..... Right arm

IWER: How compliant was R during this measurement? (Circle one)

- 1..... R was fully compliant
- 2..... R was prevented from fully complying due to illness, pain, or other symptoms or discomforts
- 3..... R was not fully compliant, but no obvious reason for this

IWER: What was R's position for this test? (Circle one)

- 1..... Standing
- 2..... Sitting
- 3..... Lying down

IWER: Did the R smoke, exercise, consume alcohol or food within the 30 minutes prior to completing the blood pressure test? (Circle one)

- 1..... Yes
- 5..... No
- 8..... Don't Know

BREATHING TEST

Equipment needed: Peak flow meter, Disposable mouthpiece(s) packaged in plastic baggy

“Next I’m going to ask you to perform a simple test that will measure how fast you can expel air from your lungs. It is important that you blow as hard and as fast as you can. I would like you to perform the test three times. When we are ready to begin, I’ll ask you to stand up. Take as deep a breath as possible. Open your mouth and close your lips firmly around the outside of the mouthpiece, and then blow as hard and as fast as you can into the mouthpiece. Like this...”

IWER: Demonstrate the test.

“Do you understand the directions for this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

IWER: Why didn’t R complete the breathing test? (Circle all that apply)

1 R felt it would not be safe

2 IWER felt it would not be safe

3 R refused or was not willing to complete the test

4 R tried but was unable to complete test

5 R did not understand the instructions

97 Other (Specify): _____

→ Go To Next Measurement

INSTRUCTIONS FOR ADMINISTERING THE BREATHING TEST:

- 1) Hand the R the instrument and a disposable mouthpiece and have the R firmly place the mouthpiece on the instrument.
- 2) R should hold the instrument lightly with the slot facing away from the R's hand so that the R's fingers do not obstruct the slot.
- 3) Ask R to stand up and take a deep breath and then place lips around the outside of the mouthpiece.
- 4) Ask the R to blow as hard and as fast as he or she can.
- 5) Record the value indicated by the marker in the chart below.
- 6) Reset the marker and repeat for a total of three tries.
- 7) Allow 30 seconds between tries.

IWER: Record measurements in chart: (Record 30 if less than 60; Record 890 if past last tick mark; Record 993 if R tried but was unable; or Record 999 if R chose not to do it.)

Measurement #	Measurement Reading
1	
2	
3	

IWER: How much effort did R give to this test? (Circle one)

1R gave full effort

2R was prevented from giving full effort by illness, pain, or other symptoms or discomforts

3R did not appear to give full effort, but no obvious reason for this

IWER: What was R's position for this test? (Circle one)

1Standing

2Sitting

3Lying down

HAND STRENGTH

Equipment needed: Dynamometer

“Now I would like to assess the strength of your hand in a gripping action. I will ask you to squeeze this handle as hard as you can, just for a couple of seconds and then let go. I will take alternately two measurements from your right and your left hands.”

IWER: Demonstrate the test.

“Before we begin, I’d like to make sure it is safe for you to do this test. Have you had surgery or experienced any swelling, inflammation, severe pain, or injury in one or both hands within the last 6 months?” (IWER: Circle one)

- 1 YES → continue with the next question
- 5 NO → skip the next question

“In which hand (have you had surgery or experienced any swelling, inflammation, severe pain, or injury in the last 6 months)?” (IWER: Circle one)

- 1 BOTH HANDS → do not complete this measure and answer question in box below
- 2 LEFT HAND ONLY → continue but do not perform test on left hand
- 3 RIGHT HAND ONLY → continue but do not perform test on right hand

“Do you understand the directions for this test?” (IWER: Circle one)

- 1 YES → continue
- 5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

- 1 YES → continue
- 5 NO → do not complete this measure and answer question in box below

IWER: Why didn’t R complete the hand strength test? (Circle all that apply)

- 1 R felt it would not be safe
- 2 IWER felt it would not be safe
- 3 R refused or was not willing to complete the test
- 4 R tried but was unable to complete test
- 5 R did not understand the instructions
- 6 R had surgery, injury, swelling, inflammation or severe pain on both hands
- 97 Other (Specify): _____

→ Go To Next Measurement

INSTRUCTIONS FOR ADMINISTERING THE HAND STRENGTH TEST:

1) “Which is your dominant hand?” (IWER: Circle one)

1Right hand

2Left hand

3Both hands equally dominant

2) Suggest R removes rings or other hand jewelry.

3) Position the respondent correctly, standing with arm at side.

4) Adjust dynamometer to hand size by turning the lever.

5) Reset arrow at zero.

6) Explain the procedure once again.

7) Let respondent have a practice with their dominant hand.

8) Reset the marker and repeat for a total of two tries.

9) Record measurements to **the nearest 0.5 kilogram** in the table below. (e.g., 10.5kg)

IWER: Record measurements in chart:

Measurement #	Left Hand	Right Hand
1	_____.____kg	_____.____kg
2	_____.____kg	_____.____kg

IWER: Record 993 if R tried but was unable. Record 999 if R chose not to do it. → Go to next measurement.

IWER: How much effort did R give to this test? (Circle one)

1R gave full effort

2R was prevented from giving full effort by illness, pain, or other symptoms or discomforts

3R did not appear to give full effort, but no obvious reason for this

IWER: What was R’s position for this test? (Circle one)

1Standing

2Sitting

3Lying down

IWER: Did the R rest their arm on a support while performing the test? (Circle one)

1Yes

5No

SEMI-TANDEM – ALL RESPONDENTS

Equipment needed: Stopwatch

“I would now like you to try to stand in different positions. I will first describe and show each position to you. Then I’d like you to try to do it. If you cannot do a particular position, or if you feel it would be unsafe to try to do it, tell me and we’ll move on to the next one. Let me emphasize that I do not want you to try to do any activity that you feel might be unsafe.”

“For the first one, I want you to try to stand with the side of the heel of one foot touching the big toe of the other foot for about 10 seconds. You may put either foot in front, whichever is more comfortable for you. Like this...”

IWER: Demonstrate the test.

“Before we begin, do you have any problems from recent surgery, injury or other health conditions that might prevent you from standing up from a chair and balancing?” (IWER: Circle one)

1 YES → discuss with him/her whether s/he should attempt each test given his/her physical problems after describing each test. Do not assume a respondent is too physically limited to attempt a test without discussing it with him/her.

5 NO → continue

“Do you understand the directions for this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

IWER: Why didn’t R complete the semi-tandem stand? (Circle all that apply)

1 R felt it would not be safe

2 IWER felt it would not be safe

3 R refused or was not willing to complete the test

4 R tried but was unable to complete test

5 R did not understand the instructions

6 R had surgery, injury or other health condition that prevented R from standing

97 Other (Specify): _____

→ Go To Side-by-Side on page 9

INSTRUCTIONS FOR SEMI-TANDEM:



- 1) Ensure R is wearing appropriate footwear (shoes with very low or no heel).
- 2) Ensure floor is level, preferably has no carpet or low-pile carpet.
- 3) Ask the R to stand up.
- 4) Stand to the side of the respondent.
- 5) Instruct the R to try to stand with the side of the heel of one foot touching the big toe of the other foot for about 10 seconds.
- 6) Instruct the R that he/she may put either foot in front, whichever is more comfortable for him/her.
- 7) Instruct the R that he/she may use his/her arms, bend his/her knees or move the body to maintain balance, but try not to move his/her feet.
- 8) If necessary, provide gentle support to the respondent's arm to help him/her get into the semi-tandem position.
- 9) Instruct the R to try to hold this position until you tell the R to stop.
- 10) Let go of R's arm. Say **"Ready, begin."** and start the stop watch immediately.
- 11) Stop the stopwatch and say **"Stop"** after 10 seconds or when the participant steps out of position or grabs your arm.
- 12) Answer the questions in the box below. If the participant is unable to hold the position for 10 seconds, record the time in seconds to two decimal places in the box below.

IWER: Did R hold semi-tandem stand for a full 10 seconds without stepping out of place or grabbing hold of anything? (Circle one)

1 Yes

5 No →

993 R tried but was unable

999 R chose not to do it

IWER: Did R use any compensatory movements of his/her trunk, arms or legs to steady him/herself during semi-tandem stand? (Circle one)

1 Yes

5 No

8 Don't Know

IWER:

- If R **was able** to complete the semi-tandem for the full 10 seconds without stepping out of place or grabbing a hold of anything → Go To Full-Tandem on page 11.

- If R **was not able** to complete the semi-tandem for the full 10 seconds without stepping out of place or grabbing a hold of anything → Go To Side-By-Side on page 9.

SIDE-BY-SIDE

Equipment needed: Stopwatch

”Now I will show you the next movement. I want you to try to stand with your feet together, side-by-side for about 10 seconds. You may use your arms, bend your knees, or move your body to maintain your balance, but try not to move your feet. Try to hold this position until I tell you to stop. Like this...”

IWER: Demonstrate the test.

“Do you understand the directions for this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

IWER: Why didn't R complete the Side-by-Side stand? (Circle all that apply)

1 R felt it would not be safe

2 IWER felt it would not be safe

3 R refused or was not willing to complete the test

4 R tried but was unable to complete test

5 R did not understand the instructions

6 R had surgery, injury or other health condition that prevented R from standing

97 Other (Specify): _____

→ Go To Walking Speed on page 13

INSTRUCTIONS FOR SIDE-BY-SIDE:

- 1) Ensure R is wearing appropriate footwear (shoes with very low or no heel).
- 2) Ensure floor is level, preferably has no carpet or low-pile carpet.
- 3) Ask the R to stand up.
- 4) Stand to the side of the respondent.
- 5) Instruct the R to try to stand with feet together, side-by-side for about 10 seconds.
- 6) Instruct the R that he/she may use his/her arms, bend his/her knees or move the body to maintain balance, but try not to move his/her feet.
- 7) If necessary, provide gentle support to the respondent's arm to help him/her get into the side-by-side position.
- 8) Instruct the R to try to hold this position until you tell the R to stop.
- 9) Let go of R's arm. Say **"Ready, begin."** and start the stop watch immediately.
- 10) Stop the stopwatch and say **"Stop"** after 10 seconds or when the participant steps out of position or grabs your arm.
- 11) Answer the questions in the box below. If the participant is unable to hold the position for 10 seconds, record the time in seconds to two decimal places in the box below.



IWER: Did R hold side-by-side stand for a full 10 seconds without stepping out of place or grabbing hold of anything? (Circle one)

1Yes

5No →

993R tried but was unable

999R chose not to do it

IWER: Did R use any compensatory movements of his/her trunk, arms or legs to steady him/herself during side-by-side stand? (Circle one)

1Yes

5No

8Don't Know

IWER: Record the type of floor surface that the balance measures were conducted on. (Circle one)

1Linoleum/tile/wood

2Low-pile carpet

3High-pile carpet

4Concrete

5Not sure

97Other (Specify): _____

IWER: How compliant was R during the balance measurements? (Circle one)

1R was fully compliant

2R was prevented from fully complying due to illness, pain, or other symptoms or discomforts

3R was not fully compliant, but no obvious reason for this

→ Go to Walking Speed on page 13

FULL-TANDEM

Record eligible time from cover

Tandem Time: ____ 30 sec ____ 60 sec

Equipment needed: Stopwatch

IWER: Refer to inside the front cover of this booklet to locate the full-tandem time for which R is eligible.

1 30 seconds full-tandem balance test

5 60 seconds full-tandem balance test

“Now I want you to try to stand with the heel of one foot in front of and touching the toes of the other foot for about [30/60] seconds. You may put either foot in front, whichever is more comfortable for you. You may use your arms, bend your knees, or move your body to maintain your balance, but try not to move your feet. Try to hold this position until I tell you to stop. Like this...”

IWER: Demonstrate the test.

“Do you understand the directions for this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

IWER: Why didn't R complete the full-tandem stand? (Circle all that apply)

1 R felt it would not be safe

2 IWER felt it would not be safe

3 R refused or was not willing to complete the test

4 R tried but was unable to complete test

5 R did not understand the instructions

6 R had surgery, injury or other health condition that prevented R from standing

97 Other (Specify): _____

→ Go To Next Measurement

INSTRUCTIONS FOR FULL-TANDEM:

Record eligible time from cover

Tandem Time: ____ 30 sec ____ 60 sec

- 1) Ensure R is wearing appropriate footwear (shoes with very low or no heel).
- 2) Ensure floor is level, preferably has no carpet or low-pile carpet.
- 3) Ask the R to stand up.
- 4) Stand to the side of the respondent.
- 5) Instruct the R to try to stand with the heel of one foot in front of and touching the toes of the other foot for about [30/60] seconds.
- 6) Instruct the R that he/she may use his/her arms, bend his/her knees or move the body to maintain balance, but try not to move his/her feet.
- 7) If necessary, provide gentle support to the respondent's arm to help him/her get into the full-tandem position.
- 8) Instruct the R to try to hold this position until you tell the R to stop.
- 9) Let go of R's arm. Say **"Ready, begin."** and start the stop watch immediately.
- 10) Stop the stopwatch and say **"Stop"** after [30/60] seconds or when the participant steps out of position or grabs your arm.
- 11) Answer the questions in the box below. If the participant is unable to hold the position for [30/60] seconds, record the time in seconds to two decimal places in the box below.



IWER: Did R hold full-tandem stand for a full [30/60] seconds without stepping out of place or grabbing hold of anything? (Circle one)

1 Yes

5 No → Enter amount of time R held stand in seconds to two decimal places ____.

993 R tried but was unable

999 R chose not to do it

IWER: Did R use any compensatory movements of his/her trunk, arms or legs to steady him/herself during full- tandem stand? (Circle one)

1 Yes

5 No

8 Don't Know

IWER: Record the type of floor surface that the balance measures were conducted on. (Circle one)

1 Linoleum/tile/wood

2 Low-pile carpet

3 High-pile carpet

4 Concrete

5 Not sure

97 Other (Specify): _____

IWER: How compliant was R during the balance measurements? (Circle one)

1 R was fully compliant

2 R was prevented from fully complying due to illness, pain, or other symptoms or discomforts

3 R did not appear to be fully compliant, but no obvious reason for this

WALKING SPEED

Is R eligible for Walking Speed Test?

_____ YES _____ NO

Equipment needed: Tape measure, Stopwatch, Masking Tape

“Next, I would like to test whether you can walk a very short distance comfortably (using a walking stick or other aid if necessary). First, I want to make sure it is safe to carry out the test. Do you have any problems from recent surgery, injury, or other health conditions that might prevent you from walking?”

(IWER: Circle one)

- 1 No apparent restriction → continue
- 2 Yes, recent surgery → do not complete this measure and answer question in box below
- 3 Yes, injury → do not complete this measure and answer question in box below
- 4 Yes, other health condition → do not complete this measure and answer question in box below

“Now let’s find a place where we can conduct the test. We will need a clear space about 12 feet long in a non-carpeted area, if possible. I’m going to place the measuring tape alongside the space where the walk will take place.”

IWER: Set up the course (98.5 Inches)

“This is our walking course. I am going to time you as you walk the course. I will be asking you to walk the course two times. I’ll walk along side you the whole time during the test. Now I’d like to demonstrate how to do the test. You will start by lining your feet up at the starting point.”

IWER: Demonstrate the test.

“Do you understand the directions for this test?” (IWER: Circle one)

- 1 YES → continue
- 5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

- 1 YES → continue
- 5 NO → do not complete this measure and answer question in box below

IWER: Why didn’t R complete the walking speed test? (Circle all that apply)

- 1 R felt it would not be safe
- 2 IWER felt it would not be safe
- 3 R refused or was not willing to complete the test
- 4 R tried but was unable to complete test
- 5 R did not understand the instructions
- 6 R had surgery, injury, or other health condition that prevented R from walking
- 7 No suitable space available

97 Other (Specify): _____

→ Go To Next Measurement

INSTRUCTIONS FOR THE WALKING SPEED TEST:

Is R eligible for Walking Speed Test? _____ YES _____ NO

- 1) Ensure that the respondent is wearing appropriate footwear at this point. If not, ask R to change shoes or to put shoes on.
- 2) Instruct R to stand with his/her feet lined up to the front of the tape.
- 3) Say the following to the R ***“I am going to time you as you walk the course at your normal pace. I will be asking you to walk the course at your usual pace a total of two times. I’ll walk along side you the whole time during the test. I’d like you to stand here with your feet lined up. Start walking when I say ‘Begin’. Walk all the way past the other end of the tape before you stop. Are you ready to go now? Begin.”***
- 4) Start stop watch when either foot is placed down on the floor across the start line.
- 5) Walk behind and slightly to the side of the R.
- 6) Stop the stopwatch when R’s foot crosses the tape and touches the floor. Record the time in chart below.
- 7) Repeat the measurement: ***“Now I want you to repeat the walk. Remember to walk at your usual pace and go all the way past the other end of the course. I’d like you to stand here with your feet lined up. Start walking when I say ‘Begin’. Are you ready to go now? Begin.”***

IWER: Record measurements in chart: (Record 993 if R tried but was unable. Record 999 if R chose not to do it)

Measurement #	Walking Speed Time
1	_____.____ (seconds)
2	_____.____ (seconds)

EX: 1 0.1 5 (seconds)

IWER: Record type of floor surface. (Circle one)

- 1Linoleum/tile/wood
- 2Low-pile carpet
- 3High-pile carpet
- 4Concrete
- 5Not sure
- 97Other (Specify): _____

IWER: Record type of aid used. (Circle one)

- 1None
- 2Walking stick or cane
- 3Elbow crutches
- 4Walking frame
- 97Other (Specify): _____

IWER: How compliant was R during this measurement? (Circle one)

- 1R was fully compliant
- 2R was prevented from fully complying due to illness, pain, or other symptoms or discomforts
- 3R was not fully compliant, but no obvious reason for this

HEIGHT

Equipment needed: Tape measure, Rafter's square, Self-adhesive note, Pencil

“Next, I would like to measure your height. To complete this measurement, I’ll be asking you to take off your shoes and stand up against a wall. I will then place this rafter’s square on top of your head and mark your height on this post-it note.”

“Do you understand the directions for this measurement?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

IWER: Why weren't you able to measure R's height? (Circle all that apply)

1 R felt it would not be safe

2 IWER felt it would not be safe

3 R refused or was not willing to complete the measurement

4 R tried but was unable to complete measurement

5 R did not understand the instructions

6 Respondent too tall, interviewer could not reach

7 There was no suitable space available

97 Other (Specify): _____

→ Go To Next Measurement

INSTRUCTIONS FOR MEASURING HEIGHT:

- 1) Ask R for location to conduct measurement.
- 2) Make sure R is standing up as straight as comfortable with his/her heels and shoulders as close to the wall as comfortable.
- 3) Place a self-adhesive note on the wall near the top of the R's head.
- 4) Position your rafter's square against the wall directly over the R's head. It should be parallel to the floor, resting on the R's head but not pressing on it.
- 5) Make a mark on the self-adhesive note at the bottom of the rafter's square.
- 6) Ask the R to move away from the wall.
- 7) Position the tape measure under a door jam, floor molding or your foot and measure from the floor to the mark on the self-adhesive note.
- 8) Record R's height in **inches** (rounded to the **nearest quarter inch**) on self-adhesive note on the wall and remove it from the wall.

IWER: Record measurement in chart:

Measurement #	Measurement
1	_____.____ inches

EXAMPLE:

Record 6 7 . 2 5 inches

Do NOT Record 5 ft 7 1/4 inches

IWER: Record the type of floor surface (Circle one)

1Linoleum/tile/wood

2Low-pile carpet

3High-pile carpet

4Concrete

5Not sure

97Other (Specify): _____

IWER: Was R wearing shoes during the measurement? (Circle one)

1Yes

5No

IWER: How compliant was R during this measurement? (Circle one)

1R was fully compliant

2R was prevented from fully complying due to illness, pain, or other symptoms or discomforts

3R was not fully compliant, but no obvious reason for this

WEIGHT

Is R eligible for Weight Measurement?

_____ YES _____ NO

Equipment needed: Scale

“Now, I’d like to measure your weight. To measure your weight, I will ask you to stand on this scale, with your shoes off, while I read the display.”

“Do you understand the directions for this measurement?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

IWER: Why weren’t you able to measure R’s weight? (Circle all that apply)

1 R felt it would not be safe

2 IWER felt it would not be safe

3 R refused or was not willing to complete the measurement

4 R tried but was unable to complete measurement

5 R did not understand the instructions

6 R is not able to stand

97 Other (Specify): _____

→ Go To Next Measurement

INSTRUCTIONS FOR MEASURING WEIGHT:

Is R eligible for Weight Measurement? _____ YES _____ NO
--

- 1) Ask R for location to conduct measurement – a hard-surface floor or thin pile carpet.
- 2) Make sure R's shoes are off or ask him/her to remove shoes.
- 3) Ask R to remove heavy objects from pockets and/or heavy sweaters as needed.
- 4) Position scale so you can see display while R is standing on it.
- 5) Turn scale on, tap middle of scale with foot, and wait for 000.0 to appear.
- 6) Ask R to stand on scale.
- 7) Record R's weight (to the nearest 1/2 pound).
- 8) R can sit down and put shoes back on.

IWER: Record measurement in chart: (Enter 993 if R tried but received an error message.)

Measurement #	Weight Measurement
1	____ _ . ____ pounds

IWER: Record the type of floor surface (Circle one)

1Linoleum/tile/wood

2Low-pile carpet

3High-pile carpet

4Concrete

5Not sure

97Other (Specify): _____

IWER: Was R wearing shoes during the measurement? (Circle one)

1Yes

5No

IWER: How compliant was R during this measurement? (Circle one)

1R was fully compliant

2R was prevented from fully complying due to illness, pain, or other symptoms or discomforts

3R was not fully compliant, but no obvious reason for this

WAIST

Equipment needed: Soft Tape measure

“Next I’m going to ask you to perform a simple measurement of your waist circumference. For this measurement it is important for you to be standing. I will ask you to identify where on your body your navel (belly button) is located. I will then ask you to place this cloth measuring tape around your waist, over your clothing, holding it securely at the level of your navel. Once the tape measure is placed appropriately around your waist then we are ready to begin. I will ask you to take a normal breath and exhale, holding your breath at the end of the exhale. I will then record the measurement.”

IWER: Demonstrate the measurement.

IWER: Ask if necessary:

“Before we begin, do you feel you are able to stand while we do this measurement?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

“Do you understand the directions for this measurement?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

IWER: Why weren’t you able to measure R’s waist? (Circle all that apply)

1 R felt it would not be safe

2 IWER felt it would not be safe

3 R refused or was not willing to complete the measurement

4 R tried but was unable to complete measurement

5 R did not understand the instructions

6 R is not able to stand

97 Other (Specify): _____

→ Go To Next Measurement

INSTRUCTIONS FOR MEASURING WAIST CIRCUMFERENCE:

- 1) Instruct R to remove bulky clothing.
- 2) R should be in the standing position and measuring tape should be applied over the clothing around the waist at the level of the navel.
- 3) Ask Respondent to point to his/her navel.
- 4) Instruct R to place the tape around his/her waist at the level of the navel. The R should hold tape in place at the navel. The tape should be snug but not tight. If the R is not able to put the tape around his/her waist, you may have them hold one end of the tape measure at their navel, and walk around the R with the other end.
- 5) Check that the tape is horizontal all away around the R.
- 6) Instruct R to take a normal breath and exhale, holding breath at end of exhale.
- 7) Read the measurement on the tape after the R exhales. Record measurement to nearest **quarter inch** in the chart below (e.g. 42.25 inches).

IWER: Record measurements in chart: (Enter 993 if R tried but was unable to do it.
Enter 999 if R chose not to do it.)

Measurement #	Waist Measurement
1	_____.____ inches

EXAMPLE:

___ 4 2 . 2 5 inches

IWER: What difficulties occurred during this measurement? (Circle all that apply)

- 1None
- 2R had breathing difficulties
- 3R was unable to hold breath at the end of the exhale
- 4R was prevented from giving full effort by illness, pain, or other symptoms or discomforts
- 5R did not appear to give full effort, but no obvious reason for this
- 97Other (Specify): _____

IWER: How compliant was R during this measurement? (Circle one)

- 1R was fully compliant
- 2R was prevented from fully complying due to illness, pain, or other symptoms or discomforts
- 3R was not fully compliant, but no obvious reason for this

IWER: Who completed the measurement? (Circle one)

- 1R completed the measurement
- 2IWER completed the measurement

IWER: Was the R wearing bulky clothing during this measurement? (Circle one)

- 1Yes
- 5No

IWER: You must have a signed consent form from the Respondent before beginning the saliva collection.

“The next measure involves collecting a saliva sample. This simply involves swishing some mouthwash in your mouth and spitting the mouthwash into a small container. Before we begin, I would like to have you read and sign this consent form.”

--

HEALTH AND RETIREMENT STUDY 2006

SALIVA COLLECTION CONSENT FORM

I have already agreed to participate in the Health and Retirement Study funded by the National Institute on Aging and conducted by the University of Michigan. The study will help provide a very complete picture of how people 50 years of age and older living in the United States are faring as they head into retirement and enjoy their retirement years.

In addition to completing the questionnaire, I am now being asked to provide a saliva sample. DNA will be extracted from the saliva and stored in a repository for future analysis. Over the course of the study, all participants will be asked to provide salivary DNA. Such analyses will allow researchers to better understand the connections between genetics and other indicators of interest such as health, economic and employment status.

I understand that if I agree to participate it means the following:

- a) I understand that I will be asked to provide a saliva sample collected by swishing a small quantity (10 ml) of Scope mouthwash in my mouth for 45 seconds. I will then be asked to spit the contents into a small container that will be sealed and packaged.
- b) I understand that the saliva sample will be sent to a repository wherein DNA will be extracted from the sample and stored for HRS-related analyses in the future. Samples are stored without my name or other identifying information. My sample will be given a unique number.
- c) I understand that I will complete these measurements only if I agree to do so. My participation in this research is completely voluntary. I am NOT required to do this, and I can stop my participation at any time. If at any time I would like to have my sample removed from storage, I can contact the Survey Director in writing and request that my sample be removed from storage and destroyed.
- d) There is no additional incentive for completing this measure, nor is there a penalty if I choose not to complete this component.
- e) The results of the extraction, any future analyses and storage information will be kept strictly confidential in the same way as the rest of the interview data. I will not be identified in any reports on this study. Records will be kept confidential to the extent provided by federal, state, and local law. However, the Institutional Review Board, the sponsor of the study (i.e. NIH, FDA, etc.), or university and government officials responsible for monitoring this study may inspect these records.

TURN OVER

- f) A Certificate of Confidentiality has been obtained from the Department of Health and Human Services for this study. With this Certificate, the investigators cannot be forced (for example by court subpoena) to disclose information that may identify me in any federal, state, or local civil, criminal, administrative, legislative, or other proceedings. Disclosure will be necessary, however, upon request of DHHS for audit or program evaluation purposes.
- g) I will be informed if the quantity of the sample provided is insufficient for storage or analyses.
- h) It will take approximately three (3) minutes to complete this measurement.
- i) There are no known risks associated with doing this interview or with participation in this study. In the unlikely event of any injury resulting from the research, no reimbursement, compensation, or free medical treatment is offered by any of the co-sponsors of this research project.
- j) I understand that although I will not receive direct benefit from my participation, others may ultimately benefit from the knowledge obtained in this study.
- k) I may ask questions about this procedure at any time and can expect truthful answers. I can ask the interviewer, or contact the Survey Director, Heidi Guyer, at the University of Michigan: (800) 759-7947.
- l) I will receive a copy of this form.

Please print name: _____

Please sign here and date: _____ **Date:** _____

Interviewer's signature: _____ **Date:** _____

Should you have questions regarding your rights as a research participant, please contact the Health Sciences Institutional Review Board, Kate Keever, 540 East Liberty Street, Suite 202, Ann Arbor, MI, 48104-2210, 734-936-0933, email: irbhsbs@umich.edu.

SALIVA SAMPLE

Equipment needed: Saliva Collection Kit (15 ml collection vial, mouthwash, 2 sheets of absorbent material, medicine measuring cup), stopwatch, padded mailing envelope

IWER: Before beginning the saliva collection measurement, has the respondent signed the consent form? (Circle one)

1 YES - Consent signed → continue

5 NO - Consent not signed → Do not complete this measure. Answer the question in the box below.

IWER: Did the R eat, drink, smoke or brush his/her teeth after starting the physical measures? (Circle one)

1 YES

5 NO

“As I mentioned, I would like you to swish some mouthwash in your mouth and spit the mouthwash into a small container.”

“Do you understand the directions for this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

IWER: Why didn't R complete the saliva sample? (Circle all that apply)

1 R felt it would not be safe

2 IWER felt it would not be safe

3 R refused or was not willing to complete the measurement

4 R tried but was unable to complete measurement

5 R did not understand the instructions

6 R ate, drank, smoked, or brushed teeth in last 30 minutes

97 Other (Specify): _____

→ Go To Blood Consent Instructions on page 26

INSTRUCTIONS FOR SALIVA SAMPLE COLLECTION:

- 1) Confirm the saliva barcode ID on top of the consent form, the Saliva Authorization Form, and the collection container.
- 2) Put on a pair of latex gloves and open the clear plastic bag.
- 3) Place one sheet of paper towel or absorbent material on the respondent's table or other hard surface where the collection will be taking place (to prevent leaving any spills).
- 4) Place the bottle of mouthwash, collection container, and medicine measuring cup on the absorbent material.
- 5) Open the mouthwash bottle and the collection container.
- 6) Pour the mouthwash into the medicine measuring cup up to the 10 ml mark.
- 7) Instruct the R to pour the mouthwash into his/her mouth and swish vigorously for 45 seconds without swallowing. Use the stopwatch to ensure that the time is met.
- 8) Instruct the R to spit the mouthwash into the collection container.
- 9) Close the collection container by screwing the lid on until it is tight.
- 10) Record the date and time on the Saliva Authorization Form.
- 11) Place the unused absorbent materials around the collection vial (to protect in the mailing process).
- 12) Place the collection container wrapped in the absorbent material and the top copy of the Saliva Authorization Form in the plastic bag. Expel the excess air from the bag and then seal the plastic bag.
- 13) Place the sealed bag in a padded envelope.
- 14) Mail the specimen as soon as possible to the lab (if not the same day as the collection, store sample in a refrigerator).

IWER: If you were not able to collect a saliva sample record the reason in the box on the previous page.

IWER: Did the R fill the vial up to the marked line? (Circle one)

1Yes

5No

IWER: What, if any, problems occurred? (Circle all that apply)

1.....None

2Scope spilled before administering

3R did not swish for the full 45 seconds

4Collection container leaked

97Other (Specify): _____

IWER: How compliant was R during this measurement? (Circle one)

1R was fully compliant

2R was prevented from fully complying due to illness, pain, or other symptoms or discomforts

3R was not fully compliant, but no obvious reason for this

IWER: Before beginning the blood spot collection you must have a signed consent form from the Respondent.

“I’d like to collect a small sample of your blood using just a finger prick. We would be very grateful if you would agree to provide us with a sample of blood. This is an important part of the study, as the analysis of blood samples will tell us a lot about the health of the population. Before we begin, I would like to have you read and sign this consent form.”

Attach Blood Spot
Barcode Label Here

Confirm Blood Spot
Barcode ID

HEALTH AND RETIREMENT STUDY 2006 BLOOD SPOT COLLECTION CONSENT FORM

I have already agreed to participate in the Health and Retirement Study funded by the National Institute on Aging and conducted by the University of Michigan. The study will help provide a very complete picture of how people 50 years of age and older living in the United States are faring as they head into retirement and enjoy their retirement years.

In addition to completing the questionnaire, I am now being asked to provide a small quantity of blood obtained via a finger prick. The blood spots will be analyzed for indicators of cardiovascular health including measures of cholesterol and blood sugar. Remaining samples will be stored in a repository for future analysis. Such analyses will allow researchers to better understand the connections between cardiovascular risk factors and other indicators of interest such as health, economic and employment status. Over the course of the study, all participants will be asked to provide blood spots.

I understand that if I agree to participate it means the following:

- a) I understand that I will be asked to provide a small blood sample for this analysis. My finger will be pricked and my blood will be collected on a blood spot collection card.
- b) I understand that the blood spot collection card will be sent to a laboratory to be analyzed for cholesterol (total cholesterol and HDL), blood sugar (Hemoglobin A1C) and C-Reactive protein. Any remaining samples will be stored for HRS-related analysis in the future.
- c) I understand that I will complete these measurements only if I agree to do so. My participation in this research is completely voluntary. I am NOT required to do this, and I can stop my participation at any time. If at any time I would like to have my sample removed from storage, I can contact the Survey Director in writing and request that my sample be removed from storage and destroyed.
- d) There is no additional incentive for completing this measure, nor is there a penalty if I choose not to complete this component.
- e) The results of the blood analyses will be kept strictly confidential in the same way as the rest of the interview data. I will not be identified in any reports on this study. Records will be kept confidential to the extent provided by federal, state, and local law. However, the Institutional Review Board, the sponsor of the study (i.e. NIH, FDA, etc.), or university and government officials responsible for monitoring this study may inspect these records.

TURN OVER

- f) A Certificate of Confidentiality has been obtained from the Department of Health and Human Services for this study. With this Certificate, the investigators cannot be forced (for example by court subpoena) to disclose information that may identify me in any federal, state, or local civil, criminal, administrative, legislative, or other proceedings. Disclosure will be necessary, however, upon request of DHHS for audit or program evaluation purposes.
- g) I will be sent a copy of my cholesterol and blood sugar results, as well as an indication if they are outside the normal range. I will be instructed to share this information with my doctor. I will be informed if the quantity of the sample provided is insufficient for analyses.
- h) It will take approximately five (5) minutes to complete this measurement.
- i) There are no known risks associated with doing this interview or with participation in this study. I may experience a small amount of pain at the time of the finger prick or at the incision site. In the unlikely event of any injury resulting from the research, no reimbursement, compensation, or free medical treatment is offered by any of the co-sponsors of this research project.
- j) I understand that although I will not receive direct benefit from my participation, others may ultimately benefit from the knowledge obtained in this study.
- k) I may ask questions about this procedure at any time and can expect truthful answers. I can ask the interviewer, or contact the Survey Director, Heidi Guyer, at the University of Michigan: (800) 759-7947.
- l) I will receive a copy of this form.

Please print name: _____

Please sign here and date: _____ **Date:** _____

Interviewer's Signature: _____ **Date:** _____

Should you have questions regarding your rights as a research participant, please contact the Health Sciences Institutional Review Board, Kate Keever, 540 East Liberty Street, Suite 202, Ann Arbor, MI, 48104-2210, 734-936-0933, email: irbhsbs@umich.edu.

BLOOD SAMPLE COLLECTION

Equipment needed: Cholesterol and A1c Collection Kit (Lab Authorization Form; 2 Blood Collection Cards, 2 Foil Blood Sample Return Bags with Desiccant; Lancets; Alcohol Prep Pad; Sterile Gauze Pad; Adhesive Bandage; Pre-addressed, Prepaid Mailing Envelope)

IWER: Did the R sign the Blood Sample consent form? (Circle one)

1 YES - Consent signed → continue

5 NO - Consent not signed → Do not complete this measure. Answer the question in the box below.

“As I mentioned, I’d like to collect a small sample of your blood. I will clean your finger with an alcohol wipe, then using a small instrument I will prick your finger and collect enough blood to fill at least four small circles on a collection card.”

“Do you understand the directions for this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

“Do you feel it would be safe for you to do this test?” (IWER: Circle one)

1 YES → continue

5 NO → do not complete this measure and answer question in box below

IWER: Why didn’t R complete the blood sample? (Circle all that apply)

1 R felt it would not be safe

2 IWER felt it would not be safe

3 R refused or was not willing to complete the measurement

4 R tried but was unable to complete measurement

5 R did not understand the instructions

6 R has hemophilia or is on medication that thins the blood (anticoagulant)

97 Other (Specify): _____

→ Thank the R and enter the requested information in the computer

INSTRUCTIONS FOR BLOOD SAMPLE COLLECTION:

- 1) Remove all parts of the Blood Collection Kit and place on a hard, clean, and dry surface (If any item is opened or broken, do not use).
- 2) Ask R to rub hands together or massage them so blood is flowing to finger tips.
- 3) Place the barcode label from the BioSafe Authorization Form at the top of the Blood Spot Collection consent form; place a barcode label from BioSafe Authorization Form on **both** Blood Collection Cards.
- 4) Write the date and time of the blood collection on the lab authorization form, and in this booklet (chart below).
- 5) Put on a pair of latex gloves.
- 6) Clean the R's finger with the alcohol prep pad and make sure the R's finger is dry before proceeding.
- 7) Remove yellow part from lancet in order to activate it. While holding the R's hand firmly, place the lancet on the side of the pad of the R's middle or ring finger.
- 8) Prick the finger with the lancet. If necessary, instruct the R to gently squeeze his/her finger from the base several times to form a large drop of blood.
- 9) With the sterile gauze pad, wipe away the first drop of blood.
- 10) When the next large drop of blood is formed, gently touch the drip to the center of one of the circles printed on the blood collection card, starting with the left-most circle. If a single drop of blood does not completely fill a circle, you may add additional drops beside the first until the circle is filled. Try not to overlap the drops. If the blood should "pool" on the surface of the card, gently tap the card to break the surface tension of the blood and allow it to flow through to the card below.
- 11) Once the R completely fills one of the circles, assist the R with filling the other circle on the card and then fill the two spots on the second card for storage.
- 12) Provide the R with a bandage.
- 13) Place or ask the R to place the used gauze pad, alcohol wipe, wrappers and lancet in the plastic bag and discard in the garbage.
- 14) Allow the blood sample to air dry for 10 to 15 minutes.
- 15) Tear off the blood sample foil return bag at the notch and open the bag.
- 16) Insert the blood collection cards into their bags and reseal the bags. Note: The desiccant pack must remain in the bag.
- 17) Place both sealed foil return bags and the top portion of the lab authorization form into BioSafe's prepaid mailing envelope.
- 18) Mail the envelope to BioSafe as soon as possible.

IWER: Record date and time of specimen collection in chart:

Specimen #	Date	Time
1		am/pm

IWER: If you were not able to collect a blood sample record the reason in the box on the previous page.

IWER: What, if any, problems occurred during the collection of the blood sample? (Circle all that apply)

1None

2R became light-headed, fainted, or nauseous

3R had difficulty getting finger to stop bleeding

97Other (Specify): _____

IWER: Who pricked the R's finger(s)? (Circle one)

1R pricked own finger(s)

2IWER pricked R's finger(s)

3Both R and Iwer pricked R's finger(s)

4Someone else pricked R's finger(s) (Specify relationship to R): _____

IWER: How many circles were filled on the first card? (Circle one) 0 1 2 3

IWER: How many circles were filled on the storage card? (Circle one) 0 1 2 3

IWER: How many times did the R's finger need to be pricked in order to fill the circles? (Circle one)

1One time

2Two times

3Two times but the R was unable to fill all four circles with blood

IWER: How compliant was R during this measurement? (Circle one)

1R was fully compliant

2R was prevented from fully complying due to illness, pain, or other symptoms or
discomforts

3R was not fully compliant, but no obvious reason for this

“Thank you for your cooperation. This concludes the physical measure and biomarkers portion of the interview. Before we continue with the interview, I need a moment to enter a few pieces of information in the computer.”

IWER: Be sure that labels have been placed on all of the forms and samples and that the samples are in the mailing envelopes. Complete the checklist on the back of this booklet. You will be prompted to enter some information now, before you proceed with the interview, but most information will be entered after you have left the respondent's house.

PHYSICAL MEASURES & BIOMARKERS CHECKLIST

Saliva Collection:

- ☐ Confirm saliva barcode ID on the consent form, Saliva Authorization Form and collection container

After sample is collected:

- ☐ Collection container, wrapped in absorbent material, and the **top** (white) copy of the Saliva Authorization Form are placed in plastic bag
- ☐ Plastic bag is sealed, placed in bubble packet and mailed in as soon as possible (or refrigerated)

Blood Spot Collection:

- ☐ Barcode label from BioSafe Authorization Form is affixed to the top of the Blood Spot Collection consent form
- ☐ Barcode label from BioSafe Authorization Form is affixed to **both** Blood Collection Cards

After sample is collected:

- ☐ Blood collection cards are air dried for 10-15 minutes
- ☐ Blood collection cards are placed in their own foil bags with desiccant pack and sealed
- ☐ Both foil bags and **top** portion of BioSafe Authorization Form are placed in prepaid mailing envelope and mailed as soon as possible

Forms to Respondent:

- ☐ 1 copy of each of the 3 consent forms
- ☐ Health Assessment Pamphlet
- ☐ High Blood Pressure Card (if necessary)

Forms to mail to AA:

- ☐ Physical Measures Booklet with signed consent forms
- ☐ Yellow copy of Saliva Authorization Form
- ☐ Bottom portion of BioSafe Authorization Form
- ☐ Completed transmittal form with the Saliva Barcode ID and the Blood Barcode ID written in

Forms to mail to Labs:

- ☐ Top (white) copy of Saliva Authorization Form
- ☐ Top portion of BioSafe Authorization Form